Group Project: Securing a Small Office Network

Cybersecurity is a rapidly growing field that offers multiple career paths and globally recognized certifications. Organizations need skilled professionals to protect their networks, systems, and data. To enter or grow in this field, certifications like **CompTIA Security+**, **Certified Ethical Hacker (CEH)**, and **Certified Information Systems Security Professional (CISSP)** play an important role. These certifications validate skills related to security fundamentals, ethical hacking, and advanced security management.

1. Career Paths in Cybersecurity

Cybersecurity provides various job roles based on skill level and specialization. Some important roles are:

a) Security Analyst

Monitors networks, analyzes logs, identifies threats, and responds to security incidents.

b) Ethical Hacker / Penetration Tester

Performs authorized hacking on networks and applications to find vulnerabilities before attackers do.

c) Security Engineer

Designs and deploys secure network architectures, firewalls, IDS/IPS, VPNs, and other security technologies.

d) Security Administrator

Manages day-to-day security operations like access control, patch updates, and monitoring tools.

e) Security Consultant

Advises organizations on best practices, policies, and solutions to improve security posture.

f) Chief Information Security Officer (CISO)

A top-level executive who develops security policies, risk management strategies, and compliance requirements.

2. Important Cybersecurity Certifications

a) CompTIA Security+

- Entry-level certification.
- Covers basic security concepts: threats, vulnerabilities, networking security, risk management, cryptography, and incident response.
- Suitable for beginners and freshers.

b) CEH (Certified Ethical Hacker)

- Intermediate level certification offered by EC-Council.
- Focuses on hacking methodologies, tools, and techniques.
- Topics include footprinting, scanning, enumeration, malware, SQL injection, wireless attacks, and exploit techniques.
- Suitable for becoming a penetration tester or ethical hacker.

c) CISSP

- Advanced certification offered by ISC².
- Suitable for professionals with experience.
- Covers 8 domains like security architecture, risk management, operations, cryptography, and network security.
- Required for roles like security manager, analyst, and CISO.

3. Group Project: Securing a Small Office Network

To secure a small office, a systematic security plan is required to ensure confidentiality, integrity, and availability of systems. The following steps are commonly implemented:

a) Network Design

- Create a secure topology with segmentation (LAN, Server Network, Guest Network).
- Use a firewall at the perimeter to control incoming and outgoing traffic.

b) Device Hardening

- Change default passwords, update firmware, and disable unused services.
- Apply operating system patches regularly.

c) Firewall & IDS/IPS Setup

- Configure firewall rules (allow only required ports).
- Use Intrusion Detection/Prevention Systems to identify attacks like port scanning and brute force.

d) Secure Wi-Fi Configuration

- Use WPA3 encryption.
- Separate guest Wi-Fi from internal office Wi-Fi.
- Hide SSID and enable MAC filtering.

e) Access Control

- Use role-based access control (RBAC).
- Implement MFA (Multi-Factor Authentication).
- Provide least privilege to employees.

f) Backup and Disaster Recovery

- Schedule automated backups.
- Use cloud + local backup strategy.
- Create a disaster recovery plan for emergencies.

g) Monitoring & Logging

- Configure log monitoring tools like SIEM.
- Track user activity, network traffic, login attempts, and system errors.

h) Security Policies & Employee Training

- Create password policy, email policy, and acceptable use policy.
- Educate employees on phishing, malware, and safe internet use.

Internet
  |
[Router / Firewall] (OpenWRT or NAT VM)
  | \
Switch   Guest AP (Guest VLAN)
 | \
Clients VLAN (192.168.10.0/24)
Server VLAN (192.168.20.0/24) <-- internal files/web/log server
Attacker VM (isolated)

Step 1: Network Scanning (Using Nmap)

This shows how attackers scan a network.
 You will demonstrate this to show why firewall + IDS is needed.

1.1 Scan Entire Network

sudo nmap -sn 192.168.1.0/24

1.2 Scan Open Ports of a Target

sudo nmap -sV <target-ip>

1.3 Vulnerability Scan

sudo nmap --script vuln <target-ip>

Step 2: Configure Local Firewall (Using UFW on target machine)
sudo ufw default deny incoming

```
sudo ufw default allow outgoing
sudo ufw allow 22
sudo ufw allow 80
sudo ufw enable
```

Step 3: Install & Run IDS (Snort) on Kali

3.1 Install Snort

```
sudo apt install snort -y
```

3.2 Check interface name

```
ip a
```

3.3 Start Snort in IDS Mode

```
sudo snort -A console -i eth0 -c /etc/snort/snort.conf
```

If your interface is *wlan0*, use that.

3.4 Perform an attack to test IDS

Run this from another terminal:

```
sudo nmap -sS <victim-ip>
```

Your Snort window should show ALERT messages.

Step 4: Secure Wi-Fi (Theory + Demo Using Commands)

Even if your Kali VM is wired/NAT, you can show Wi-Fi security demo.

Check Wi-Fi networks

```
sudo iwlist wlan0 scan
```

Explain types of security (WPA2, WPA3).

Step 5: Access Control Demonstration

On a Linux target (Ubuntu):

Create users & permissions

```
sudo adduser user1
sudo adduser hrmanager
sudo groupadd hr
sudo usermod -aG hr hrmanager
sudo mkdir /hr-files
sudo chown :hr /hr-files
sudo chmod 770 /hr-files
```

Step 6: Monitoring & Logging from Kali

Monitor live packets (Wireshark)

Open:

```
sudo wireshark &
```

Or using tcpdump:

```
sudo tcpdump -i eth0
```

Step 7: Backup (Simulation)

On target (Ubuntu):

Backup using rsync

```
rsync -av /home/user/ /home/backup/
```

FINAL DEMONSTRATION FLOW

1. Scan the network using nmap (Kali)

→ show hosts, open ports.

2. Harden server using firewall

→ show blocked ports.

3. Run Snort IDS on Kali

→ show alerts when scanning.

4. Show Wi-Fi security info

→ explain WPA2/WPA3.

5. Show access control

→ file permission & user groups.

6. Show logs / monitoring

→ tcpdump / Wireshark capture.

7. Show backup

→ rsync command.

This completes the full **Small Office Network Security Practical**.

Emerging Threats: IoT, AI, Quantum Security

1. IoT (Internet of Things) Security Threats

IoT devices include smart cameras, smart lights, smart TVs, smart watches, medical sensors, etc.

Why IoT is risky

- Devices have weak passwords
- Not updated regularly
- Limited security features
- Always connected to internet

Major IoT Threats

1. Botnets
   Hackers infect IoT devices and use them for DDoS attacks (e.g., Mirai botnet).

2. Unauthorized Access
   Weak/default passwords allow attackers to control cameras, routers, or smart appliances.

3. Data Theft
   IoT devices collect personal info → easy target for hackers.

4. Device Hijacking
   Attackers can control devices (e.g., turning off sensors in industries).

5. Firmware Attacks
   Outdated firmware allows malware to remain inside IoT devices.

Impact

- Privacy loss
- Smart home hacking
- Industrial shutdown
- Healthcare threats (hacking pacemakers, pumps)

2. AI (Artificial Intelligence) Security Threats

AI improves security but also increases new cyber threats.

Why AI is a threat

- Hackers use AI to automate attacks
- AI can generate malware or fake identities
- AI models can be tricked or manipulated

Major AI Threats

1. Deepfakes
   Fake videos/voices used for fraud, political manipulation or scams.

2. AI-powered phishing
   AI generates realistic messages → higher success rate.

3. Adversarial Attacks
   Small changes in images or data confuse AI systems

(e.g., self-driving cars misreading stop signs).

4. Model Poisoning
   Hackers inject fake data to corrupt an AI model.

5. AI Malware
   Malware that learns and evolves to escape detection.

Impact

- Identity fraud
- Large-scale scams
- Wrong decisions by autonomous systems
- Security systems bypassed

3. Quantum Security Threats

Quantum computers can perform calculations at extremely high speed.

Why Quantum is risky

- Current encryption (RSA, ECC) can be broken by quantum computers.

Major Quantum Threats

1. Breaking Cryptography
   Quantum algorithms (Shor's algorithm) can crack RSA and ECC quickly.

2. Harvest Now, Decrypt Later
   Attackers store encrypted data now and decrypt later when quantum computers become strong enough.

3. Threat to Blockchain
   Digital signatures used in blockchain can be broken.

4. Weak Post-Quantum Readiness
   Most systems are NOT prepared for quantum threats.

Impact

- Passwords, banking, government data can be exposed
- Encrypted messages can be decrypted
- Loss of privacy and national security risks

Project Ideas:

Secure WiFi Setup: Configure WPA2, disable WPS, enable MAC filtering.

Mini Pen-Test: Scan a vulnerable VM (Metasploitable) and report findings.

Awareness Campaign: Create posters/videos on phishing prevention.

BEGINNER-FRIENDLY CYBERSECURITY PROJECT IDEAS

(Simple + good for presentations)

1. Password Strength Analyzer

- Tool checks password strength
- Shows suggestions
- Use Python + Regex

2. Wi-Fi Network Scanner

- Scan nearby networks
- Show signal strength & encryption types
- Use `iwlist`, `nmcli` or Python

3. Port Scanner

- Simple version of Nmap

- Python sockets

4. File Integrity Checker

- Detect file changes using hashing (MD5/SHA256)
- Alerts if a file was modified

5. Phishing Awareness Website

- Create a page that teaches users how phishing works
- No harmful code
- Good educational project

🚀 INTERMEDIATE CYBERSECURITY PROJECTS

(Use Kali tools, logs, IDS, automation)

6. Log Monitoring System

- Collect logs from system

- Detect suspicious login attempts
- Use `Python`, `fail2ban`, `rsyslog`

7. Simple IDS (Intrusion Detection System)

- Monitor traffic on Kali using `Scapy`
- Detect port scans or brute force attempts
- Print alerts

8. Secure Small Office Network Setup

(This matches your group project)

- Use Kali as attacker
- Ubuntu as server
- Configure firewall, IDS, backup, access control

9. Vulnerability Scanner Mini Tool

- Use Python + Nmap libraries
- Scan open ports & weak services
- Generate a small report

10. Web Application Security Testing Report

- Test a demo website (DVWA / Juice Shop)
- Document XSS, SQLi, CSRF
- Use Burp Suite / OWASP ZAP

ADVANCED CYBERSECURITY PROJECTS

(If you want something powerful for mini project/final year)

11. Machine Learning Based Malware Detection

- Dataset: benign vs malware samples (features only)
- Train ML model to classify threats

12. Network Traffic Anomaly Detection

- Capture packets with Wireshark/tcpdump
- Extract features
- Use ML to detect anomalies

13. AI-Based Phishing URL Detection

- Use ML to classify URLs as safe or phishing
- Features: length, symbols, domain age

14. Automated Patch Management Tool

- Script checks outdated packages
- Generates report
- Can auto-update

15. Quantum-Safe Encryption Demo

- Use Post-Quantum Algorithms (e.g., CRYSTALS-Kyber)
- Encrypt messages & show difference from RSA

IoT & Cloud Security Projects

16. IoT Device Security Auditor

- Scan IoT devices for open ports/weak passwords
- Show risk levels

17. Secure Smart Home Simulation

- Raspberry Pi + sensors
- Show attacks vs defenses

18. Cloud Security Monitoring Dashboard

- Use AWS free tier
- Detect login failures, IAM issues

- Show data in dashboard

DIGITAL FORENSICS PROJECTS

19. Android Forensics

- Extract data from Android backup
- Analyze messages, calls, images
- Use Autopsy, ADB tools

20. Disk Forensics Case Study

- Create a fake crime scenario
- Recover deleted files
- Generate forensics report

BEST RECOMMENDATION FOR YOU (based on your messages)

Here are top 5 projects perfect for you:

1. Securing a Small Office Network (Already working on)
2. Log Monitoring + Alert System using Linux & Python
3. Simple IDS using Python + Scapy
4. Vulnerability Scanner using Python & Nmap
5. Phishing URL Detector using Machine Learning

These are easy to explain, easy to demo, and perfect for college.